Sustainable Living Inc

Carbon Footprint and Energy Audit

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Acknowledgment

Sustainable Living Inc

Hiran Prashanth
Environmental Sustainability Auditor

15 Sep 2021

Carbon footprint and Energy audit at BABA Institute of Technology and Sciences

The Sustainable Living Inc acknowledges with thanks the cooperation extended to our team for completing the study at BABA Institute of Technology and Sciences (BITS - VIZAG).

The interactions and deliberations with BITS - VIZAG team were exemplary and the whole exercise was thoroughly a rewarding experience for us. We deeply appreciate the interest, enthusiasm, and commitment of BITS - VIZAG team towards environmental sustainability.

We are sure that the recommendations presented in this report will be implemented and the BITS - VIZAG team will further improve their environmental performance.

Kind regards,

PRINCIPAL
Baba Institute of Technology and Sciences
Bakkannapalem (V), Madhurawada (P)
Visakhapatnam

Yours sincerely,
Hiran Prashanth

Hiran Prashanth
Environmental Sustainability Auditor
Sustainable Living In

About Auditor

Hiran Prashanth is a sustainability consultant based in London. He has over 14 years of experience in climate change and environmental sustainability. He was working with the Confederation of Indian Industry (CII) before moving to London to pursue a master's degree at King's College, London.

Hiran Prashanth was awarded the 'Best Sustainability Assessor' by the Honorable Minister for HRD, Mr. Prakash Javadekar. Hiran Prashanth is a CII certified carbon footprint expert and a resource efficiency expert. He has trained more than 1000 industry personnel across the world on climate change and sustainability. He is a guest a faculty at IIM Lucknow and SIBM, Pune. His credentials can be found on Hiran Prashanth | LinkedIn. Sustainable Living Inc provides services on carbon footprint, energy audit, resource management and embodied carbon.

Executive Summary

The growth of countries across the world is leading to increased consumption of natural resources. There is an urgent need to establish environmental sustainability in every activity we do. In a modern economy, environmental sustainability will play a critical role in the very existence of an organization.

An educational institution is no different. Built environment, especially an educational institution, has a considerable footprint on the environment. Impact on the environment due to energy consumption, water usage and waste generation in an educational institute is prominent. Therefore, there is an imminent need to reduce the overall environmental footprint of the institution.

As an Institution of higher learning, BABA Institute of Technology and Sciences (BITS - VIZAG) firmly believes that there is an urgent need to address the environmental challenges and improve their environmental footprint.

True to its belief, BITS - VIZAG has solar PV for generating clean energy for its campus. 330 KWp of solar panels has been installed in the campus. BITS - VIZAG is also in the process of replacing conventional lamps with energy efficiency lamps. Sustainable Living Inc Team congratulates BITS - VIZAG team for their efforts.

Keeping BITS - VIZAG's work in energy efficiency, we recommend the following to be taken by the competent team at BITS - VIZAG:

Work towards achieving carbon neutrality: INDC emphasizes creating an additional carbon sink of 2.5 to 3 billion tonnes of CO2 equivalent through additional forest and tree cover by 2030. BITS - VIZAG's net carbon emission for the year 2020-21 is **94.20 MT CO2e**. BITS - VIZAG should focus on energy efficiency, renewable energy, and carbon sequestration as tools that will enable them to offset the present carbon emissions and achieve carbon neutrality.

Installation of biogas plant: In 2020-21, BITS - VIZAG had used 5.93 MT of LPG. There is an opportunity to install a biogas plant to generate biogas from sewage water. Presently, sewage water is being let out to the drain without treatment. An opportunity exists to generate biogas from the untreated sewage water and use the generated biogas to substitute LPG used in the college. By generating biogas from sewage water, about 0.93 MT of LPG can be replaced which will result in carbon savings of 2.79 MT CO2e.

Reduce contract demand with electricity board: Presently, the majority of the power consumed by BITS – VIZAG is generated through their solar PV system in the campus. Therefore, the energy consumption from the grid is at the minimum. The contract demand with the grid is 120 KVA. This means that the electricity board will charge BITS – VIZAG a minimum KVA charges, 80% of 120 KVA which is 96 KVA. The actual consumption of the campus is only 32 KVA. However, the college has to pay for a minimum demand charge of 96 KVA. Each KVA is charged at Rs. 475. By reducing the contract demand to 60 KVA from 120 KVA, the college will pay only Rs. 22,800 per month instead of Rs. 45, 600. BITS – VIZAG can save Rs. 2,73,600 per year.

Improve energy efficiency of the college: It is recommended to adopt latest energy efficient technologies for reducing energy consumption in fans, lighting, and air conditioners. We recommend the following projects to be implemented at the earliest:

- Replace conventional 70W ceiling fans with energy efficient BLDC fans of 30W
- Installation of Air conditioners energy savers
- Replace conventional lamps with energy efficient LED lamps

Carbon Footprint and Energy Audit

BABA Institute of Technology and Sciences (BITS - VIZAG) and Sustainable Living Inc are working together to identify opportunities for improvement in energy efficiency and carbon reduction. This report highlights all the potential proposals for improvement through the audit and analysis of the data provided by BITS - VIZAG for lighting, air conditioning, ceiling fans, and biogas potential.

The report also details the carbon emissions from college operations. For carbon emissions, scope 1 and scope 2 emissions are calculated from the data submitted by BITS - VIZAG. The report emphasizes the GHG emission reduction potential possible through a reduction in power consumption.

Submission of Documents

Carbon footprint and energy audit at BITS - VIZAG was carried out with the help of data submitted by BITS - VIZAG team. BITS - VIZAG team was responsible for collecting all the necessary data and submitting the relevant documents to Sustainable Living Inc for the study.

Carbon Footprint and Energy Audit

Data submitted and collected was used to calculate the carbon footprint of the campus and assess energy consumption and finally provide necessary recommendations for environmental improvement.

Note

Carbon footprint and energy audit are based on the data provided by BITS - VIZAG team and discussions the Sustainable Living Inc team had with BITS - VIZAG team. The scope of the study does not include the exclusive verification of various regulatory requirements related to environmental sustainability.

Sustainable Living Inc has the right to recall the study if it finds (a) major violation in meeting the environmental regulatory requirements by the location and (b) occurrence of major accidents, leading to significant damage to ecology and environment.

OPPORTUNITIES FOR IMPROVEMENT

As a part of the overall environmental improvement study at BITS - VIZAG, carbon footprint calculations were also carried out. The objective of calculating the carbon footprint of the campus is find the present level of emissions from campus operation and what initiatives that the BITS - VIZAG can take to offset the emissions. By offsetting the emissions, the college can become carbon neutral in the future by adopting energy efficient processes, increase in renewable energy share and tree plantation.

Carbon footprint calculations:

To help delineate direct and indirect emission sources, improve transparency, and provide utility for different types of organizations and different types of climate policies and business goals, three "scopes" (scope 1, scope 2, and scope 3) are defined for GHG accounting and reporting purposes.

For calculating carbon footprint of the campus, Scope 1 & Scope 2 emissions are being considered. Since day scholars use college provided transportation and hostelers stay in campus, Scope 1 and Scope 2 are the highest contributor to overall emissions. For this reason, Scope 3 is not being calculated.

Scope 1: Direct GHG Emissions

Direct GHG emissions occur from sources that are owned or controlled by the company, for example, emissions from combustion in owned or controlled DG sets, canteen, vehicles, etc.; emissions from chemical production in owned or controlled process equipment. Direct CO2 emissions from the combustion of biomass shall not be included in scope 1 but reported separately.

BITS - VIZAG Scope 1 emissions for 2020-21:

Sources of Scope 1 emissions in BITS - VIZAG:

1) LPG used for canteen

2) Diesel used for generator

S No	Fuel Type	Description	Activity Data	Units	CO_2 eq. Emissions (tons)
1	LPG	Canteen	5.93	MT	17.67
2	Diesel	Transportation	11.40	KL	30.10
3	Diesel	Generator	0.30	KL	0.79

Total Scope 1 emissions of BITS - VIZAG : 48.60 Tons (for year 2020-21)

Scope 2: Electricity Indirect GHG Emissions

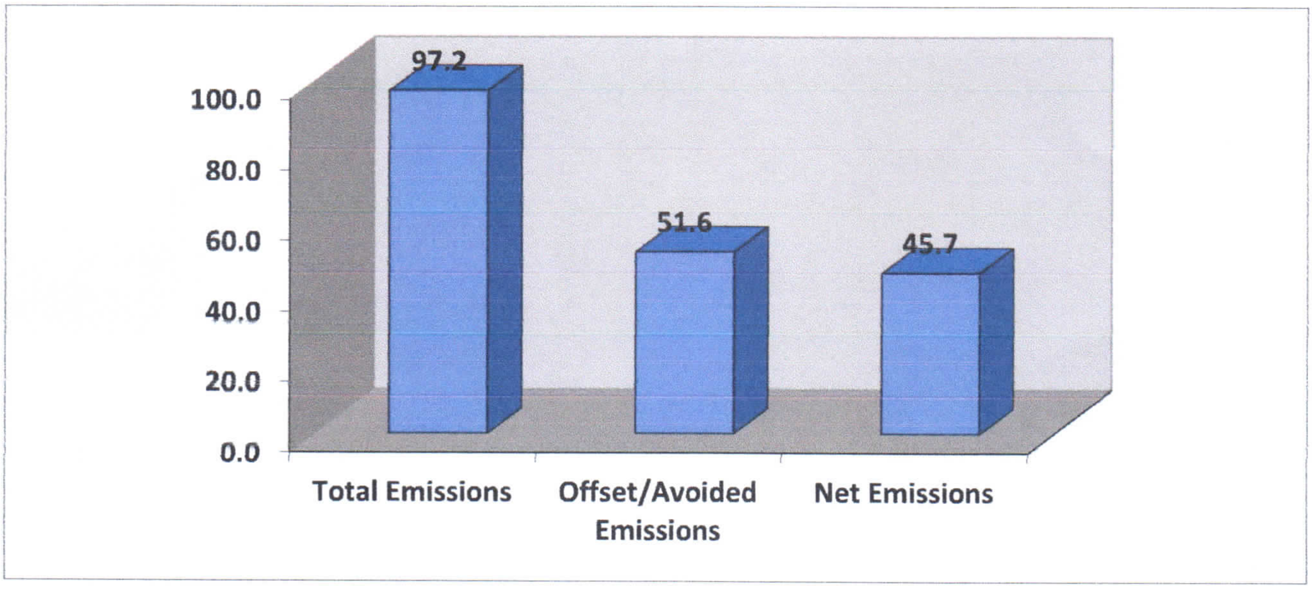
Scope 2 accounts for GHG emissions from the generation of purchased electricity consumed by a company. Purchased electricity is defined as electricity that is purchased or otherwise brought into the organizational boundary of the company. Scope 2 emissions physically occur at the facility where electricity is generated.

BITS - VIZAG Scope 2 emissions for 2020-21:

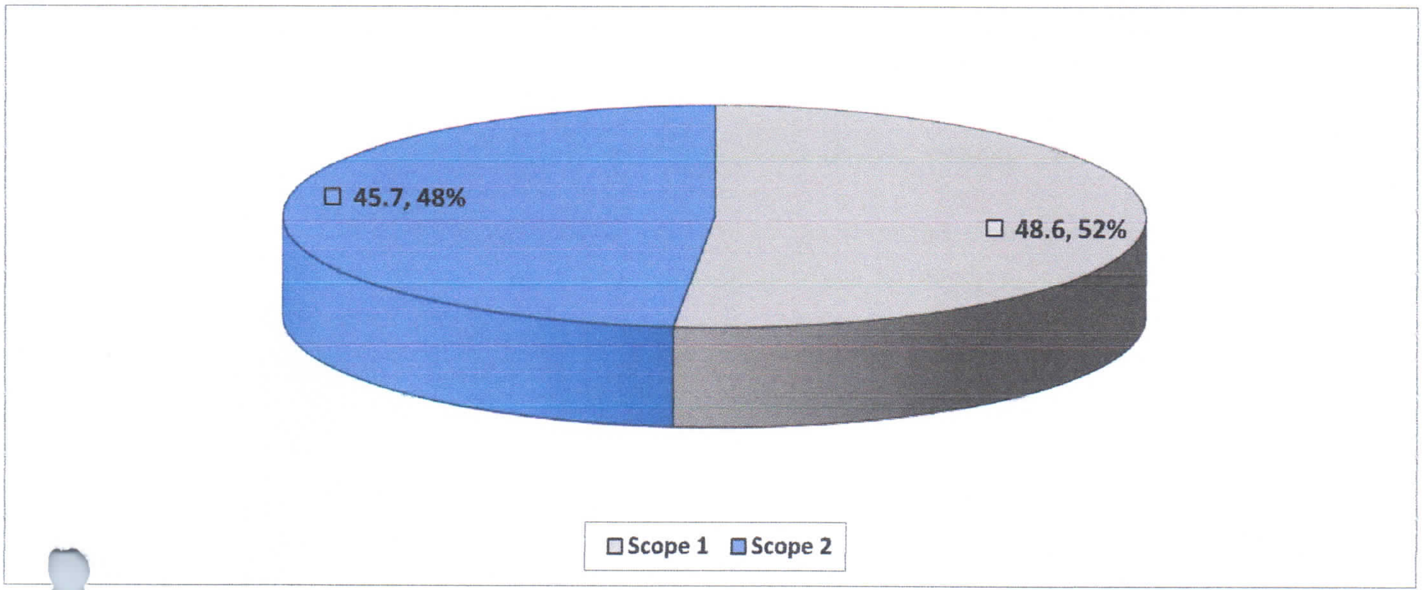
Electricity purchased from grid : 55,689

Solar energy produced : 62,904

Scope 2 Breakup



GHG Emission Summary of BITS - VIZAG



Scope 1	48.60	**MT CO2 eq.**
Scope 2	45.70	**MT CO2 eq.**
Total	94.20	**MT CO2 eq.**

Develop a roadmap to increase contribution of renewable energy in the overall energy consumption

To have a continued focus on increasing renewable energy utilization to 100% which will also lead to reduction in GHG emissions, it is suggested to develop a detailed roadmap on RE utilization. The road map should broadly feature the following aspects -

- Renewable energy potential of BITS - VIZAG and the maximum offset that can be achieved at BITS - VIZAG
- Percentage substitution with renewable energy that BITS - VIZAG wants to achieve in a specified time frame
- Key tasks that needs to be executed to achieve the renewable energy target
- Specific financial break up for each of the projects highlighting the amount required, available and the utilization status as on date
- A regular review mechanism to ensure progress along the lines of the roadmap should be framed
- The roadmap should also highlight important milestones/key tasks, anticipated bottleBITS - VIZAGks & proposed

Renewable energy roadmap should be used as a base to frame GHG emissions reduction target

It is suggested to use the developed renewable energy roadmap to correlate the GHG reduction that each of the renewable energy project will achieve. This approach will provide a base to set targets for reduction in GHG emissions. The action plan for renewable energy will shoulder the action plan for GHG emissions reduction and work towards achieving carbon neutrality.

Explore the option of other onsite and offsite renewable energy projects

The renewable energy field has been witnessing many private investors due its increased market demand and attractive policies in many states. There are Renewable Energy Independent Power Producers (RE IPPs) who have installed RE based power plants like wind, small hydro and solar PV. GOC can consider having a long-term power purchase agreement with these RE IPPs in purchasing fixed quantity of power for a period of 5 to 10 years.

Evolve a system to monitor the implementation of various GHG mitigation opportunities

BITS - VIZAG has an action plan to reduce its GHG emissions. BITS - VIZAG should also evolve a system to monitor the implementation of various GHG mitigation opportunities. It is recommended to use a Gantt chart to mark out the action plan for the activities and track its implementation. Gantt chart will serve as an excellent way to instantly monitor and comprehend all different tasks in one place which would ease tracking of implementation.

Install biogas plant at BITS - VIZAG campus

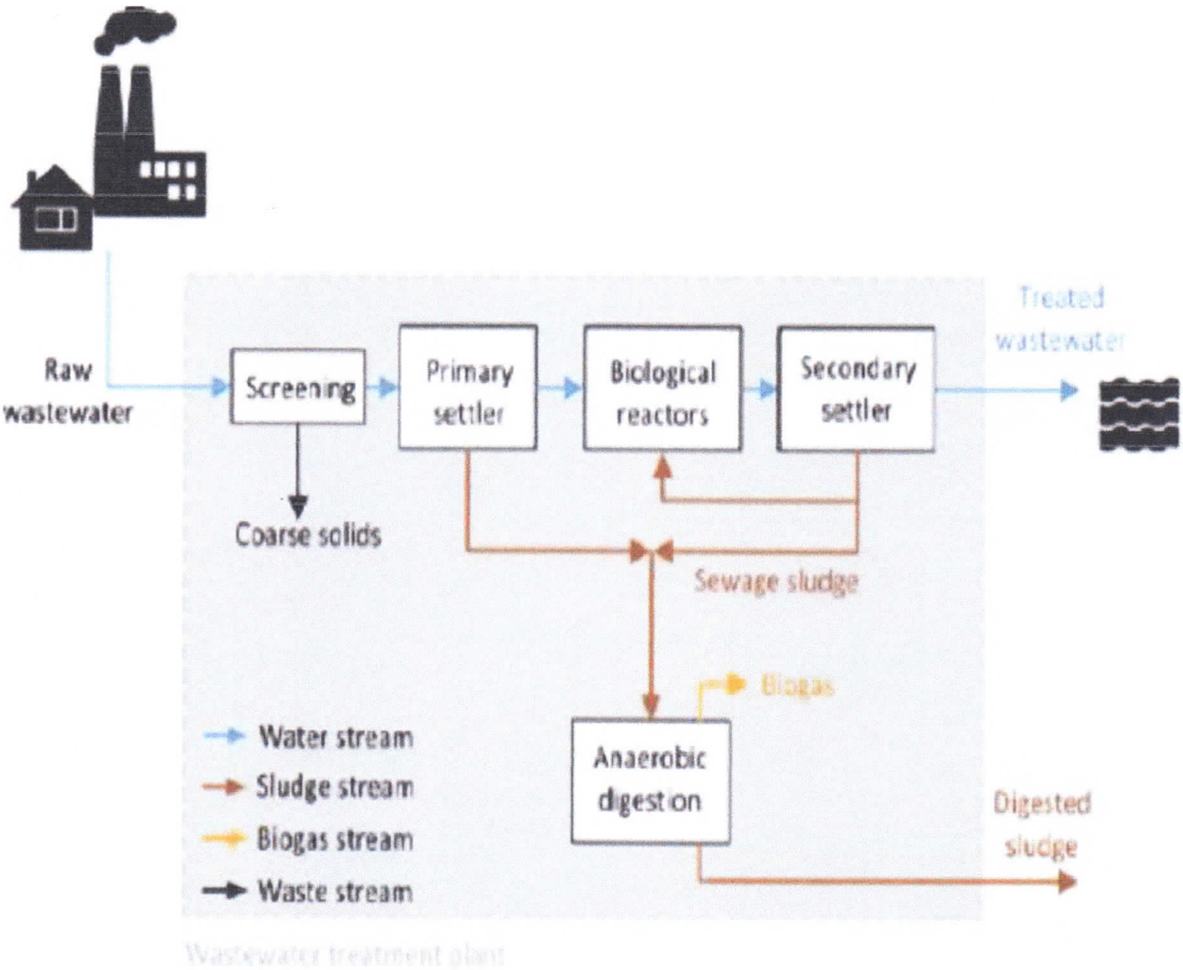
Presently, sewage water is being let out to the drain without treatment. An opportunity exists to generate biogas from the untreated sewage water and use the generated biogas to substitute LPG used in the college.

BITS - VIZAG had used 5.93 MT of LPG. By generating biogas from sewage water, about 0.93 MT of LPG can be replaced which will result in carbon savings of 2.79 MT CO2e.

Biogas Production Potential of Wastewater

The sewage water is a useful waster as 1% of it in any quantity is a sludge which when subjected to anaerobic digestion will produce biogas. Wastewater is the effluent from household, commercial establishments and institutions, hospitals, industries and so on. Sewage water source contains large amount of organic material which can be efficiently recovered in as sludge which and when subjected to anaerobic digestion, the sludge produces methane gas (biogas).

Biogas is a mixture of gases containing 50-75% Methane, and 25-50%Carbon dioxide while 0-10% Nitrogen, 0-3% Hydrogen disulphide and 0-2% Hydrogen may be present as impurities which is produced by anaerobic digestion of organic material i.e. a sequential enzymatic breakdown of biodegradable organic material (Biomass) in the absence of oxygen. The process is usually carried out in a digester tank known as biodigester. Biogas is an important energy source used as cooking gas, to generate electricity, etc. thus producing biogas from wastewater is an efficient and sustainable waste management and renewable energy technique. One of the major environmental problems of the world today is waste management and wastewater constitutes a huge environmental problem to the society thus the need for wastewater treatment to recover and also recycle the recovered water for usage.



The physical process: this is the mechanical treatment of the water that involves removal of debris from the raw wastewater right from the point it enters the plant. The screening and primary settling of debris. Wastewater enters the treatment plant through the inlet chamber from where it is channeled to the coarse screen that removes solid waste.

The biological process: this involve the biotreatment of the sewage in the bioreactors. It is the heart of the treatment plant where a biological process takes place. The bioreactors of a treatment plant are usually large tanks consisting of several mammoth rotors and submersible mixers. While the rotor introduces atmospheric oxygen into the sewage, the submersible mixers keep the biomass in suspension thus several reactions takes place in the bioreactors.

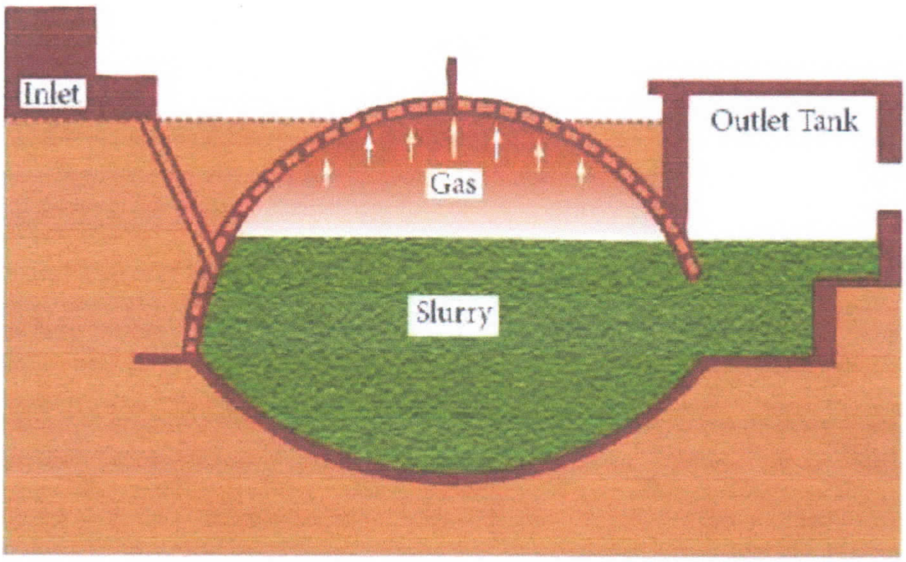
From the bioreactor, the sewage enters the sedimentation tank. Here the biological process ends and sludge is separated from water such that the clean water is passed to the disinfection tank for disinfection and onward discharge for use while the sludge is removed by the returned activation sludge (RAS) pump that removes and sends part to the anaerobic digestion chamber while some are return to the anaerobic bioreactor for reactivation.

Production of biogas is an anaerobic digestion whereby microorganisms break down biodegradable material in the absence of oxygen to produce methane/carbon dioxide used to generate electricity and heat. Sludge from the treatment plant (primary and activated sludge) is the main feedstock (biodegradable organic matter) in the biogas production plant of a wastewater treatment plant and the biogas production process involves series of steps. The combine sludge resulting from primary and secondary water treatment is gathered, sieved and thickened to a dry solids content of up to 7% before entering the digesters. Optionally, the sludge can be pretreated by disintegration technologies with the aim to improve the gas yield. In the anaerobic digestion process, the sludge is pumped into the anaerobic continuously stirred tank reactors where digestion takes place.

In the process, microorganisms break down part of the organic matter that is contained in the sludge and produce biogas, which is composed of methane, carbon dioxide and trace gases. The raw biogas produced is dried and hydrogen sulphide and other trace substances removed and burned in burners after treatment. The digested sludge is dewatered, and the water reintroduce into the treatment plant while the remaining undigested matter used for organic fertilizer.

Calculations:

Sewage water available per day	:	5 KL (Least value considered for calculation)
Sludge in 10KL of sewage water	:	1% (100 kg)
From 6kg of organic waste	:	1 kg of biogas can be produced
Therefore, from 50 kg	:	8.33 kg of biogas can be produced
Kg of biogas	:	0.45kg of LPG
Per day equivalent LPG production	:	3.25 kg per day
Annual LPG production for 250 days	:	937.50 kg
Annual emission reduction potential	:	2.79 T CO_2



ENERGY EFFICIENCY

Annual energy consumption of BITS - VIZAG campus is 55,689 units. There are major blocks in the campus which consumes energy for their operation. Major energy consumers are:

1. Fans
2. Air conditioners
3. Lighting

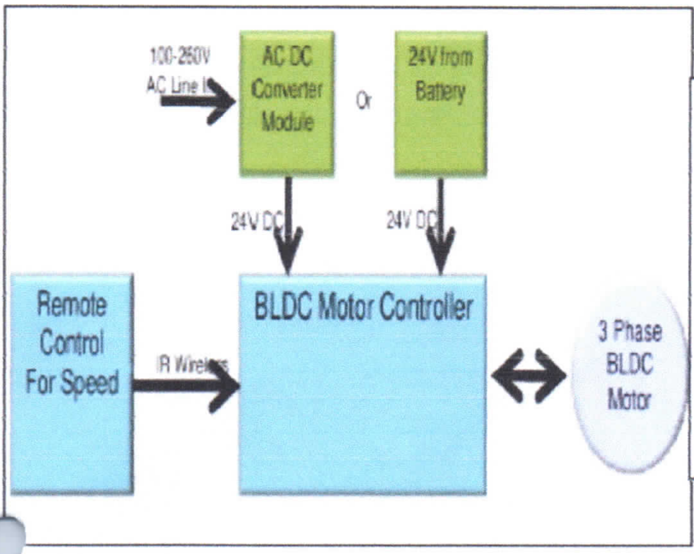
Replace Conventional Ceiling Fans with Energy Efficient BLDC Fans

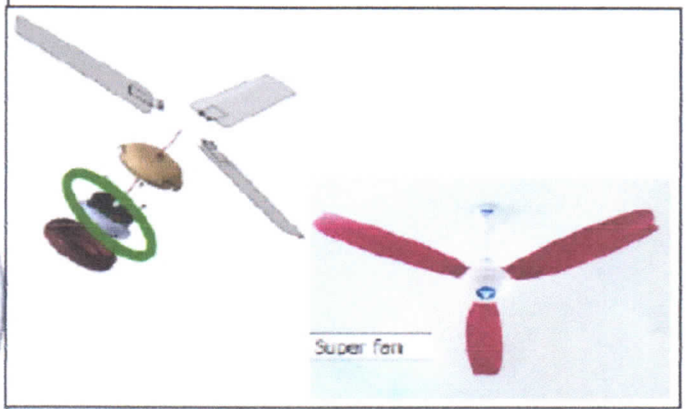
During the Energy Audit at BITS - VIZAG, a detailed study was carried out to identify the potential for replacing the existing ceiling fans with BLDC super fans. There are 457 fans operating in BITS - VIZAG campus.

Instead of conventional ceiling fans, latest technology BLDC fans which consume only 30W can be installed in the newly constructed building. A brushless DC (BLDC) motor is a synchronous electric motor powered by direct-current (DC) electricity and having an electronic commutation system, rather than a mechanical commutator and brushes. A BLDC motor has an external armature called the stator, and an internal armature called the rotor.

The rotor can usually be a permanent magnet. Typical BLDC motor-based ceiling fan has much better efficiency and excellent constant RPM control as it operates out of fixed DC voltage. The proposed BLDC motor and the control electronics operate out of 24V DC through an SMPS having input AC which can vary from 90V to 270V. The operational block diagram of a BLDC motor is as follows:

Calculations:





With the replacement of existing ceiling fans with Super Fans the energy consumption is likely to reduce by 55% per fixture. Considering 100 fans being replaced with super-efficient BLDC fans, 3.50 kW can be saved. Considering the average operating hours to be 2000 and unit cost as Rs. 7.50, the calculations are as follows:

Total no. of fans in college	:	457
Energy consumption per fan	:	70 W
Total energy consumption of fans	:	70W X 100 fans
	:	7 kW
Super-efficient BLDC fans energy consumption	:	30 W
Savings from 70W to 30 W	:	55%
Total savings in fans energy consumption	:	55% of 7kW
	:	3.5 kW
Savings per year	:	3.5 kW X 2000 hrs X Rs. 7.50 / unit
	:	Rs. 0.75 Lakhs
Investment	:	Rs. 2, 50, 000
	:	52 months
Annual emission reduction potential	:	6.00 T CO2

Replace Conventional Lamps with LED Lamps

As per the data submitted, the total number of all the lighting fixtures installed are 200 tube lights. BITS - VIZAG has already taken an excellent step of replacing conventional lamps to LED lamps. We congratulate the team for the excellent step.

Under failure replacement policy, at least 100 lamps can be changed in the first year.

Types of fixtures	**36 W Tube**
No of fixtures	100
No of hours in Operation	2000

The campus should be keen in harnessing the day lighting available thereby reducing the use of artificial lighting.

Based on the occupancy, monitoring should be ensured to reduce excessive consumption of energy.

Major savings in energy through lighting fixtures can be achieved by replacing all the above existing fixtures with LED's meeting the required LUX levels. The LED's being less energy consuming while maintaining the equivalent lux is the more sustainable option. The replacement of lighting fixtures should be done as per failure replacement policy i.e. change the old fixture with LED when it fails

Advantages of LED

- Lower energy consumption: The energy consumption of LEDs is low when compared to the other conventional sources for the same amount of Lumen output.

Performance comparison of different type lights

Type of Lamp	Lumen/ Watt	CRI	Life hours
HPSV lamps	90-120	Bad (22-25)	15,000-20,000
Metal Halide lamps	65-00	Good (65-90)	18,000
LED lamps	100-150	Very Good (> 80)	10,000 – 12,000

- **High S/P ratio**: LEDs have higher scotopic/photopic ratio (S/P ratio). The eye has two primary light sensing cells called rods and cones – cones function in day light and process visual information whereas rods function in night light. The cone dominated vision is called photopic and the rod dominated vision is called scotopic. The S/P ratio indicates the measure of light that excites rods compared to the light that excites cones. In office environments, illumination is more effective if the S/P ratio is high as it is under scotopic region. LEDs hence are ideally suited for these applications as they have a high S/P ratio.
- **Longer life-time**: LEDs have longer life time of around 1,00,000 hours. This is equivalent to 11years of continuous operation or 22 years of 50% operation.
- **Faster switching**: LED lights reach its brightness instantly upon switching and can frequently be switched on/off without reducing the operational life expectancy.
- **Greater durability and reliability**: As LEDs are solid-state devices and uses semi-conductor material; they are sturdier than conventional sources that use filaments or glass. LEDs can also withstand shock, extreme temperatures and vibration as they don't have fragile materials as components.
- **Good Colour Rendering Index (CRI)**: The color rendering index, i.e., measure of a light sources' ability to show objects as perceived under sunlight is high for LEDs. The CRI of natural sunlight is 100 and LEDs offer CRI of 80 and above.
- LED offers more focused light and reduced glare. Moreover, it does not contain pollutants like mercury. LED technology is highly compatible for solar lighting as low-voltage power supply is enough for LED illumination.

Calculations are as follows:

Existing Lighting Fixtures	**36 W Tube**
Existing power consumption (kW)	3.60 kW (100 lamps)
Proposed LED Wattage (W)	15
LED power consumption (kW)	1.50 kW
Energy saving (kW)	2.00 kW
Operating hours	2000

Annual monetary savings	:	Rs 38,250/-
Investment needed	:	Rs 90,000/-
Payback period	:	2.50 Years
Annual Emission reduction potential	:	4.18 MT of CO_2

Install Air conditioners energy saver for spilt air conditioners:

Present status: As per the data obtained from BITS - VIZAG team, the campus has majorly 1.5 TR units installed. There are 18 spilt air conditioners installed and operate 8 hours a day.

Recommendation:

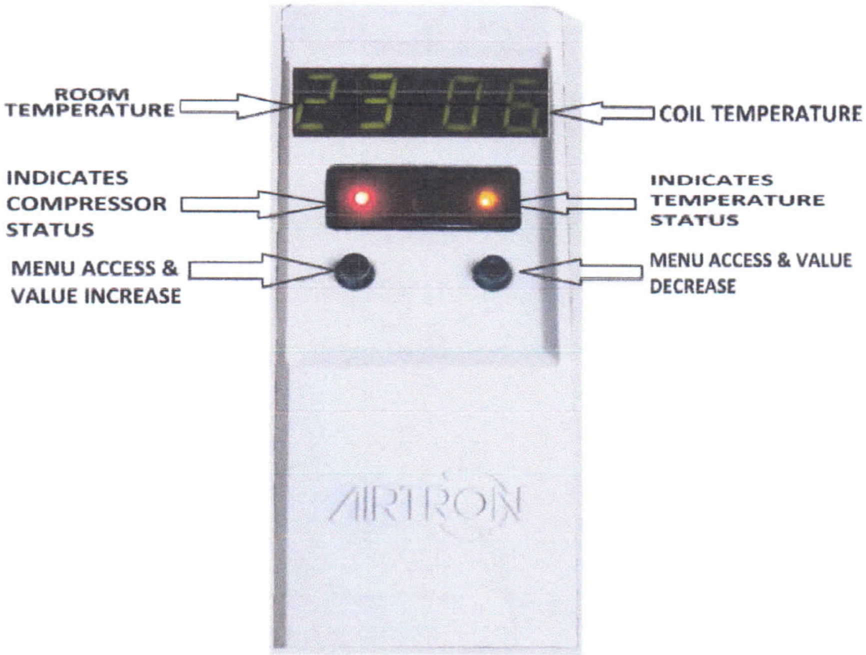
We recommend installing "Airtron", an energy saver that can be installed at every individual unit of AC. The Airtron is the world's most advanced AC SAVER, with all the controls of a Precision AC. The Airtron's dual sensors reference the Room and Coil & Ambient Temp, and uses complex, multiple algorithms in a "closed -loop circuit" to reduce the Compressor Run-Time, to ensure the high savings while maintaining and displaying the Set temperature accurately. The Airtron is Programmable for geographical location and climate and adapts automatically to changes in season and ambient conditions.

This unique device has been developed on Patent-Published technology and approved by leading MNC'S, PSU'S and Govt. Departments. The Airtron is validated by EESL (Energy Efficiency Services Ltd.), Ministry of Power, Government of India, for 44% savings. The Airtron has been validated on all AC's- Inverters, 5 Star, Splits, Multi-Splits, Packages, ducts, Windows, Cassettes from 1.0 - 20.0 TR, LG ltd, Videocon Ltd, Tata Communications, L&T, Nestle, Ashok Leyland etc. The AIRTRON comes with a Remote for setting the Room Temperature, and in a Non-Flammable Polycarbonate Enclosure, with SMPS Power Supply, to tolerate w ide Voltage and Current fluctuations, Surges, Spikes and Sags.

In our case, Airtron installation can reduce the energy consumption of each fixture by 15% on a conservative basis. For a total energy consumption, for air conditioners, as 20 units per hour, 3 units per hour can be saved. It is recommended to install Airtron energy saver in a phase wise manner preferably in the batches of 10 units.

Saving Calculation: Considering the operating hours to be 2000 and unit cost as Rs 7.50/-.

- Monetary annual savings : Rs 45,000/-
- Total investment : Rs 80,000/-
- Payback period : 22 months (2 years)
- Annual emission reduction potential : 4.92 MT CO2



Install solar water heater for hostel hot water requirements

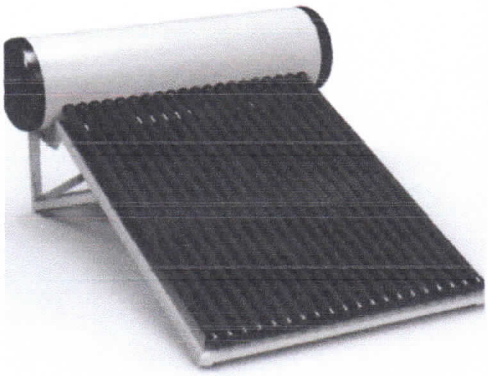
Heaters are being used for the hot water requirements of the hostel Electrical heaters are one of the highest energy consumers in the hostel with each heater consuming 800W of energy.

Replacing the electrical heaters with solar water heaters is the best solution for eliminating the power consumption of the heaters.

The following explanation of solar water heaters is taken from www.bijlibachao.com.

A solar water heater is a system that utilizes solar energy (or the energy from sunlight) to heat water. It has a system that is installed on a terrace or open space where it can get sunlight and the energy from the sun is then used to heat water and store it in an insulated tank. The system is not connected to electricity supply and thus does not have an on-off switch, but it uses the sunlight throughout the day to heat the water and store it in the storage tank. Most of the solar water heater on a sunny day can provide heater water at about 68° ±5° C temperature. Water from the storage tank can then be used for any application as desired. One can feed this heated water to the electric geyser so that when sunlight is not enough, it uses electric energy to heat the water to the desired set temperature. This is also called Hybrid Water Heater.

Solar Water Heaters Types and Benefits



Flat Plate Collectors (FPC) System	Evacuated Tube Collectors (ETC) System
Long lasting as they are metallic. But are expensive	Fragile but cheaper.
Can work in colder regions with sub zero temperature but will need an anti freeze solution.	Very good for colder regions where the temperature is sub zero.
In places with salty water a heat exchanger is required with FPC system.	Require regular cleaning where the water is salty.

Benefits of a 100 lts Solar Water Heater in India.

	Northern Region	Eastern Region	Southern Region	Western Region
Expected no. of days of use of hot water per year	200 days	200 days	300 days	250 days
Expected yearly electricity saving on full use of solar hot water (units of electricity)	1000	1000	1500	1250

Bijli Bachao!
because saving electricity saves money

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For this report, a 100-liter capacity solar water heater is considered. A 100-liter, EPC solar water require requires 20 square feet of space. The energy saving from the system is calculated a follow:

Heat required (kcal) = M (Mass of water) x Cp (Specific heat of water) x delta T (Difference in starting temperature and desired temperature)

kW saving = M (Mass of water) x Cp (Specific heat of water) x delta T (Difference in starting temperature and desired temperature) X 0.0012 (conversion from kcal to kW)

= 100 kg X 1 X (50 Deg C – 25 Deg C) X 0.0012

= 3 kW

Therefore, for heating 100 litres of water, the energy saving would be 3 kW.

Cost of 500-liter EPC solar water will be Rs. 60,000.

For a 500-litre solar water heater the energy saving will be 15 kW.

Cost saving for 250 days of operation will be Rs. 28,000.

Pay back will be in 25 months.

Conclusion

BITS - VIZAG has initiated few energy efficiency activities in their campus. While Sustainable Living Inc appreciates the plant team for their efforts, we would like to emphasize that opportunity exists further reduce the energy consumption. Installation of renewable energy is to be given major focus. RESCO model can be adopted to install renewable energy without upfront capital investment. We in Sustainable Living Inc are sure that all the recommendations mentioned in the report will be implemented by BITS - VIZAG team and the overall environmental performance of the campus will be improved.

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Visakhapatnam

List of Vendors

Equipment	Supplier Name	Contact Person	Mail Address	Contact Number
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BLDC Ceiling Fans	Atomberg Technologies Pvt Ltd	Ms Roshni Noronha	roshninoronha@atomberg.com	9987366655
BLDC Ceiling Fans	Versa Drives	Mr Sathish	sathish@versadrives.com	94885 94382
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